

Perfume Vaporiser Adaptor

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An electric perfume vaporiser slowly vaporises liquid perfume in a bottle using an electrical heating module. A wick transports the liquid from the bottle into the heating zone on top by capillary action. The heating is done at a temperature maintained within a narrow range suitable for the perfume, so that its release is neither too fast nor too slow.

This circuit helps in controlling the temperature of the liquid vaporiser's

PARTS LIST

Semiconductors:

- TRIAC1 - BT136 triac
- DIAC1 - DB3 diac
- D1 - 1N4007 rectifier diode
- LED1 - 3mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2 - 2.2-mega-ohm
- VR1 - 500-kilo-ohm potentiometer

Capacitor:

- C1 - 150nF, 400V polyester

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 2-pin connector
- S1 - On/off switch

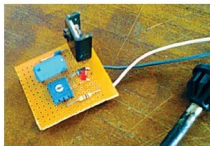


Fig. 1: Author's prototype

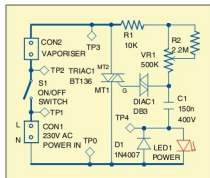


Fig. 2: Perfume vaporiser adaptor circuit

heating element and thereby the rate of vapourisation and level of fragrance delivery. As a rule, the fragrance in the bottle should last for at least 30 days. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of perfume vaporiser adaptor. It is an inexpensive solid-state heat controlling circuit built around the commonly-available triac BT139 (TRIAC1).

Phase control is one of the most common forms of triac control. A basic phase-control circuit includes resistors, a capacitor and a diac. Resistors R1 and R2, potentiometer VR1 and capacitor C1 provide a phase shifted AC signal to diac DB3 (DIAC1). When the voltage on C1 exceeds the break-over voltage of DB3, it breaks down and triggers

Test Points

Test point	Details
Across TP0 and TP1	230V AC
Across TP2 and TP3	Output depends on position of VR1
Across TP0 and TP4	Around 1.1V AC

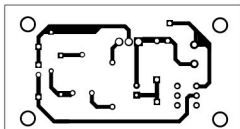


Fig. 3: An actual-size PCB for perfume vaporiser adaptor

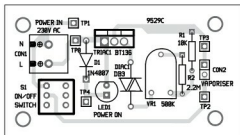


Fig. 4: Component layout for the PCB



TRIAC1. This allows the vaporiser to connect to 230V supply through TRIAC1, thus allowing the heating element to heat up.

The 500-kilo-ohm potentiometer (VR1) provides a smooth continuous control of power to the heating element of the vaporiser. By varying it, you can get different levels of fragrance output.

Working of the circuit is simple. On closing switch S1 the 3mm red LED1 glows, indicating that circuit is switched on. Turning VR1 increases or decreases the release of perfume.

This circuit can also be used as an adaptor for insect repellent vaporisers, air fresheners, etc.

Construction and testing

An actual-size, single-side PCB for the perfume vaporiser adaptor is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable plastic case. Connect the two wires from CON2 to the 2-pin plug of the vaporiser.

Fix VR1 on the front side of the cabinet and connect it to PCB using wires. Fix 2-pin connectors for power supply and the vaporiser.

A small heat sink should be provided for TRIAC1. But the heat sink should not be allowed to come in contact with any external metallic object.

Before connecting the vaporiser to the circuit, verify that the voltages at various test points are as per table.

Caution. Since the circuit is directly operated from 230V AC mains, only experienced persons should assemble it. To avoid lethal shock, take necessary precautions while handling the AC mains voltage. ●

The author is a freelance writer and circuits designer